Analytic Unitarization

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This HISQ force uses smearing to attenuate taste-exchange processes, but there are a few that are enhanced by the smear. This latter, undesirable effect can be mitigated by *reunitarization*, i.e. by projecting the smeared link back to U(3). There are many possible strategies for doing the projection. Here we discuss the "analytic" method.

Let V be the smeared link. The matrix

$$Q = V^\dagger V \tag{1}$$

is Hermitian. Then

$$W = VQ^{-1/2} \tag{2}$$

is a member of U(3). Computing the inverse root of Q is carried out analytically by leveraging the Cayley-Hamilton theorem, and correspondingly, the derivative of the reunitarization, which is needed for the force, can be worked out analytically too. For more details, see Appendices B and C of Ref. [1].

References

[1] A. Bazavov et al. Scaling studies of QCD with the dynamical highly improved staggered quark action. *Phys. Rev. D*, 82(7):074501, 2010.